



Practical No. 9 - Write Algorithm, Code and Output

Roll No. : 34	Name: Mehek Sawant
Class : SY-IT	Date/Time of Submission: 18-08-2026

Q. 1. Write algorithm for the following:

a. Write a program to count the occurrences of a specific word in a file using regular expressions.

Code:

```
import re
```

```
def count_word(filename, word):
```

```
    try:
```

```
        #open file in read mode
```

```
        with open(filename, "r") as f:
```

```
            text=f.read()
```

```
        matches= re.findall(rf'\b{word}\b', text, re.IGNORECASE)
```

```
        return len(matches)
```

```
    except FileNotFoundError:
```

```
        print("Error: File not found")
```

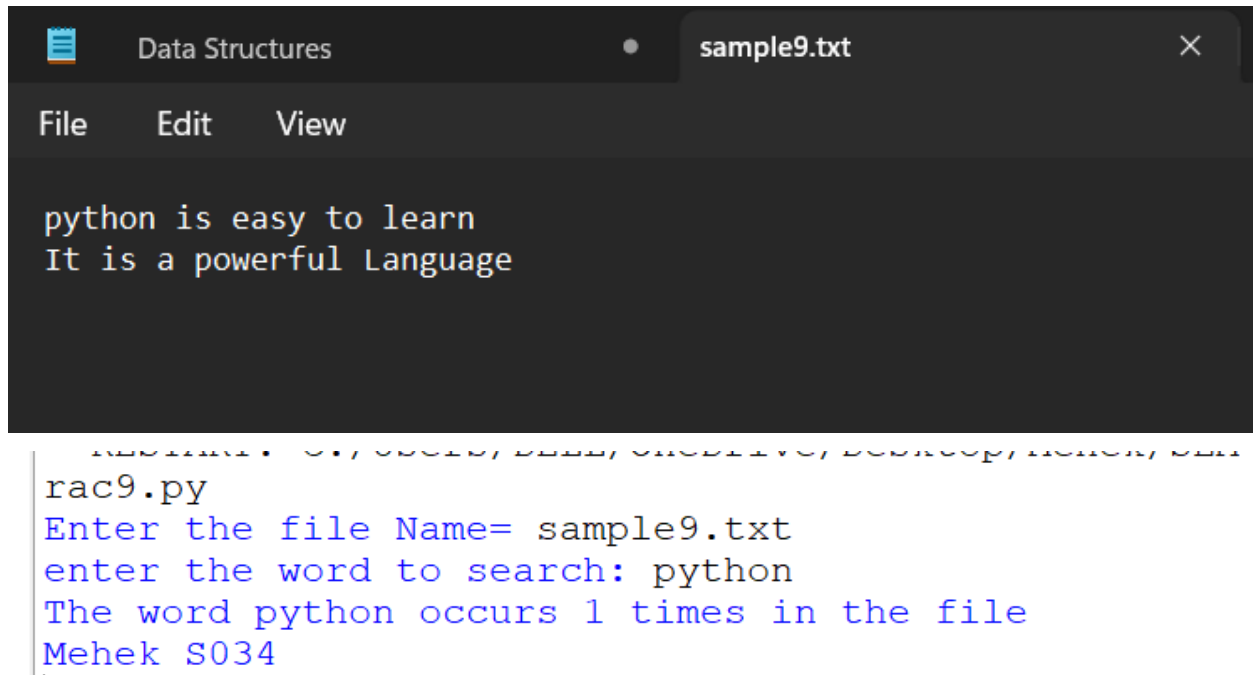
```
        return 0
```

```
filename= input("Enter the file Name= ")
```

```
word = input("enter the word to search: ")
```

```
count= count_word(filename, word)
print(f"The word {word} occurs {count} times in the file")
print("Mehek S034")
```

Output:



```
python is easy to learn
It is a powerful Language

rac9.py
Enter the file Name= sample9.txt
enter the word to search: python
The word python occurs 1 times in the file
Mehek S034
```

b. Write a program to extract all hyperlinks ([from an HTML file](#))

Code:

```
import re

def extract_hyperlinks(filename):
    try:
        # Open and read the HTML file
        with open(filename, "r", encoding="utf-8") as f:
            content = f.read()

        # Regex pattern to match href values inside <a> tags
        pattern = r'<a\s+(?:[^\>]*\s+)?href="([^\"]*)"'

        # Find all hyperlinks
        links = re.findall(pattern, content, re.IGNORECASE)
```

```

        return links

    except FileNotFoundError:
        print("Error: File not found!")
        return []

#----Main Program----
if __name__ == "__main__":
    filename = input("Enter the HTML file name: ")
    hyperlinks = extract_hyperlinks(filename)

    if hyperlinks:
        print("\nExtracted Hyperlinks:")
        for link in hyperlinks:
            print(link)
    else:
        print("No hyperlinks found or file empty.")

print("Mehek Sawant S034")

```

Output:

```

= RESTART: C:/Users/DELL/OneDrive/Desktop/Mehek/SEI
rac9b.py
Enter the HTML file name: Prac9.html

Extracted Hyperlinks:
https://www.google.com
https://www.github.com
Mehek Sawant S034

```

Q. 2.

- a. Create a text file with student names and roll numbers.
- b. Write a program to read and display all names.
- c. Write a program to count total words in the file.
- d. Append “Attendance: Present” to the file.

Code:

```
print("--Create a text file with student names and roll numbers.--")
file = open("student9.txt", "w")
file.write("101 Rahul\n")
file.write("102 Priya\n")
file.write("103 Amit\n")
file.write("104 Sneha\n")
file.write("105 Neha\n")
file.close()
```

```
print("File Created Successfully")
```

```
print("---program to read and display all names.---")
file = open("students.txt", "r")
```

```
for line in file:
    print(line)
```

```
file.close()
```

```
print("---Write a program to count total words in the file.---")
file = open("students.txt", "r")
```

```
data = file.read()
words = data.split()
```

```
print("Total words:", len(words))
file.close()
```

```
print("---Append “Attendance: Present” to the file.---")
```

```
file = open("students.txt", "a")

file.write("Attendance: Present\n")

file.close()

print("Attendance added successfully.")
```

Output:

```
= RESTART: C:/Users/DELL/OneDrive/Desktop/Mehek/SEM 3/Python.
rac9.py
--Create a text file with student names and roll numbers.--
File Created Successfully
---program to read and display all names.---
101 Rahul

102 Priya

103 Amit

104 Sneha

105 Neha

---Write a program to count total words in the file.---
Total words: 10
---Append "Attendance: Present" to the file.---
Attendance added successfully.
```